

KRUX: A Gamified, Edge-AI Powered Smart Bin Ecosystem for Decentralized Plastic Recycling

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Abstract

The global plastic crisis is exacerbated by a lack of financial incentive for individuals to recycle, coupled with high contamination rates in traditional waste streams. In this paper, we present KRUX, a decentralized, gamified Smart Bin ecosystem designed to shift the sorting process to the consumer source. KRUX integrates an ESP32-powered autonomous smart bin with a Progressive Web App (PWA) built on React and Firebase. By utilizing TensorFlow.js for on-device machine learning inference, the system accurately classifies plastic types directly on the user's mobile device, eliminating server costs and ensuring privacy. Furthermore, a secure real-time handshake protocol between the physical hardware and the web application autonomously verifies physical deposits, rewarding users with cryptocurrency-style 'KRUX Coins'. This holistic approach effectively digitizes and gamifies the circular economy.

Index Terms

Edge AI, Gamification, Smart Bin, ESP32, TensorFlow.js, Computer Vision, Circular Economy, Firebase

I. INTRODUCTION

Global plastic production exceeds 380 million tons annually, yet less than 9% is successfully recycled. The inefficiency of the current recycling pipeline stems primarily from two factors: a lack of consumer motivation and severe cross-contamination in collection bins. Traditional waste management relies on expensive downstream Material Recovery Facilities (MRFs) to sort mixed waste. To address these inefficiencies, this paper introduces **KRUX**, a low-cost, gamified recycling platform. KRUX incentivizes users by transforming the recycling process into an interactive digital experience. By decentralizing the classification and sorting process to the consumer level via an AI-powered PWA and an autonomous smart bin, KRUX ensures that recycling streams remain uncontaminated from the point of collection.

II. SOFTWARE ARCHITECTURE

The software layer of KRUX is designed as a highly accessible Progressive Web App (PWA) to ensure frictionless adoption across iOS and Android devices without requiring heavy native application downloads.

A. Tech Stack and Infrastructure

The frontend is constructed using React 18, Vite, and Tailwind CSS, ensuring a highly responsive and modern user interface. Backend operations are handled entirely by Google Firebase. Cloud Firestore manages user profiles, streaks, and transaction histories, while Firebase Authentication secures user sessions. Crucially, the Firebase Realtime Database (RTDB) acts as a zero-latency communication bridge between the web application and the physical bin.

B. On-Device Machine Learning

To eliminate the latency and recurring cloud computational costs associated with server-side AI, KRUX performs image classification entirely on the edge. Utilizing TensorFlow.js and a MobileNet architecture, the application captures the device's camera stream and executes the model against the local GPU/NPU. This architecture not only reduces operational costs to zero but also guarantees total user privacy, as no image data is ever transmitted over the network.

C. Gamification Engine and Fraud Detection

User retention is driven by a comprehensive gamification engine. Users earn 'KRUX Coins' based on the specific type of polymer recycled (e.g., PET yields higher rewards than mixed plastics). The platform features global leaderboards, achievement badges, and streak multipliers. To prevent system exploitation, multi-layered fraud detection is implemented, including rapid-scan cooldowns, minimum ML confidence thresholds, and hardware-level validation.

III. HARDWARE ARCHITECTURE

The physical KRUX Smart Bin operates autonomously, acting as the final physical validator for all digital transactions.

A. Microcontroller and Actuation

The core processing unit of the bin is an ESP32 microcontroller, selected for its integrated WiFi capabilities and low power consumption. The bin is equipped with an Infrared (IR) obstacle sensor situated at the intake chute, and a standard 180-degree servo motor responsible for mechanically routing the deposited material into the correct internal compartment.

B. Event Synchronization

When a user physically drops a scanned item, the IR sensor detects the object breaking its beam. The ESP32 calculates the necessary servo actuation to sort the plastic, and simultaneously constructs a JSON payload. This payload is transmitted via an HTTP PUT request directly to the specific bin's node on the Firebase RTDB, triggering the final transaction validation on the user's mobile device.

IV. SYSTEM INTEGRATION AND USE CASE FLOW

The seamless integration between the hardware and software layers is achieved through a secure handshake protocol without the need for a central intermediary server.

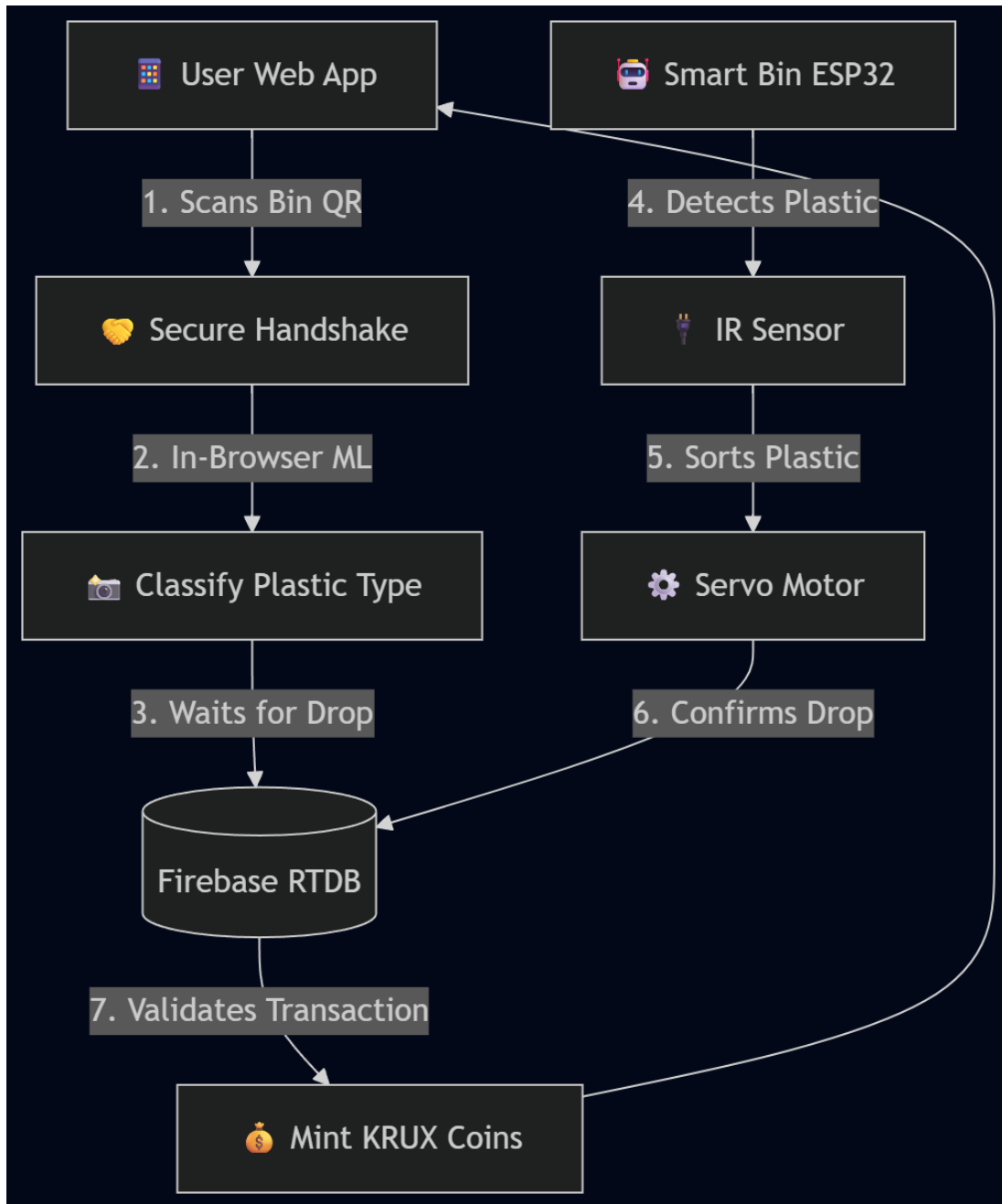


Fig. 1. KRUX System Architecture demonstrating the data flow between the React PWA, Firebase RTDB, and the ESP32 Hardware.

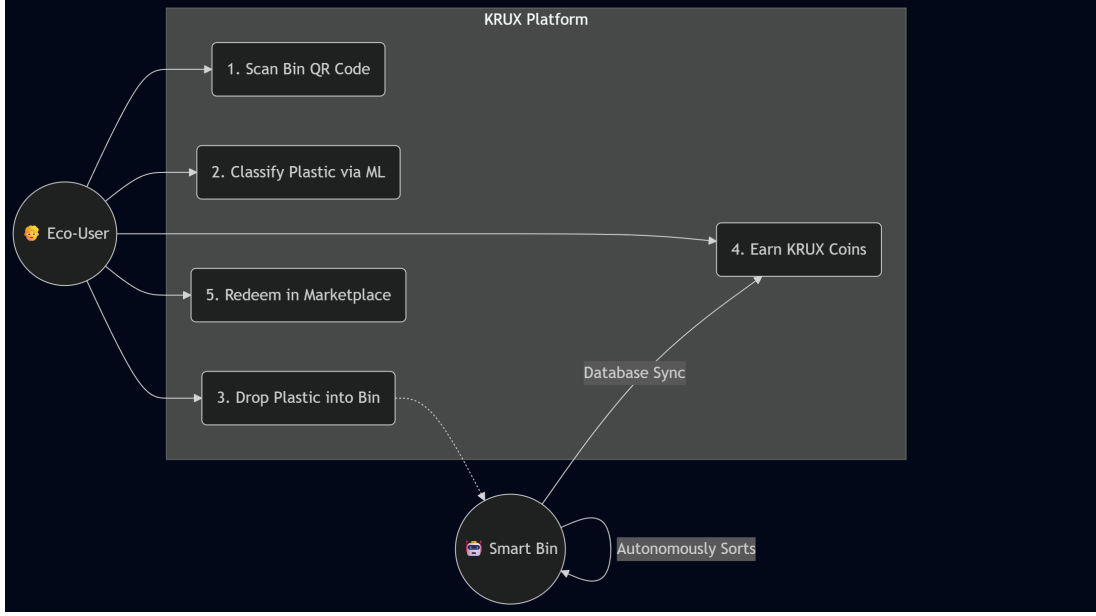


Fig. 2. Use Case Flow mapping the user journey from scanning the QR code to receiving KRUX Coins upon physical validation.

A. The Handshake Mechanism

The transaction lifecycle is defined as follows:

- 1) **Initiation:** The user scans a physical QR code on the bin, linking their session to the bin's unique ID on the RTDB.
- 2) **Classification:** The user scans the plastic item using the in-app ML model. Upon successful classification, the app enters a 30-second escrow phase.
- 3) **Physical Drop:** The user deposits the item. The ESP32 detects the drop and updates the RTDB state.
- 4) **Resolution:** The PWA detects the state change on the RTDB, terminates the escrow timer, and securely mints KRUX coins to the user's Firestore profile.

V. CONCLUSION AND FUTURE SCOPE

The KRUX ecosystem demonstrates a scalable, cost-effective approach to decentralizing waste segregation. By leveraging on-device ML and IoT synchronization, the system eliminates traditional backend bottlenecks while providing a highly engaging user experience. Future iterations aim to expand the on-device ML dataset for higher precision across multi-layered plastics. Additionally, the integration of physical pneumatic sorting mechanisms for industrial-scale deployments will be explored, further closing the loop on a fully automated, gamified circular economy.

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